

CS617 – Monroe College: Week 4 Assignment exercises: Using R

Project 2

The Airports case study for section 2

Data files

`tsa_claims2.csv`

`GlobalAirportDatabase.csv`

Scenario

Your boss liked your previous report so much he wants you to expand on it by completing the specified tasks and answering some additional questions!

Tasks and questions

- Read the data from the `tsa_claims2.csv` file, not the `tsa_claims1.csv` file.
- Clean the data:
 - Improve the column names by removing the spaces from them.
 - Drop any rows that contain NA values.
 - Store the date columns with the Date type.
 - Convert the ClaimAmount and CloseAmount columns to the double type. (*Hint: You will first need to remove the dollar signs and semicolons from the character strings.*)
- Answer the following questions:
 - If a claim is approved, what percent of the claim amount did the airports pay on average? If the claim is settled, what percent of the amount did the airports pay on average?
 - What are the five airports with the most claims?
 - Has the average amount paid out by airports increased or decreased over time?
- Create a map that shows the location of each airport in the continental U.S. and set the size of the airport's dot based on its number of claims. Get the location of each airport from the `GlobalAirportDatabase.csv` file.

Instructions

- Create a script that answers the questions shown above.
- Whenever possible, reuse code from the script for section 1 and improve it to reflect what you've learned in section 2.
- Whenever it makes sense, use a plot to answer a question.
- Use comments to indicate which parts of your script perform tasks and answer questions. If it's helpful, use comments to state your answers to questions.

